Ajay Venkatesh

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Education

New York University | *Master of Science in Computer Engineering* ***(GPA – 3.9/4)*** Sep 2024 – May 2026

Experience

Campus Mesh **Jan 2026 – Present**

Software Engineer *New York, US*

- Collaborated and maintained scalable **Node.js microservices** powering real-time academic workflows for **10K+ users**, including study groups, messaging, notifications, and profile management.
- Built a personalized **recommendation engine** using collaborative filtering and behavioral analytics, increasing user engagement by **28%** and retention by **50%**.
- Authored **LLM-based retrieval pipelines** for CampusIQ (AI Assistant), improving contextual accuracy by **30%** through semantic retrieval and structured data indexing.

Amazon **May 2025 – Aug 2025**

Software Development Engineer Intern *Seattle, US*

- Architected a secure UI console for Amazon fulfillment center teams to view image/video evidence across FCs; presented the **system design** for approval and established Role Based Access **(4+ roles)** using **OAuth, Midway SSO, and HELIS**.
- Provisioned **serverless Infrastructure-as-Code** using **AWS CDK** with API Gateway, Lambda, DynamoDB, S3, and IAM, reducing provisioning time by **70%** and enabling **multi-stage, multi-region CI/CD deployments.**
- Built a **React 18** frontend with optimized API integration, advanced filtering, client-side caching, reducing page load time by **40%**.
- Engineered backend services in **Java** using **Coral, Smithy, Dagger**, and **CBOR serialization**; **applied low-level design and design patterns** reduced backend execution time by **35%** and ensure reliability by unit and integration testing using **Mockito**.

New York University (Novel AI Technologies, Inc.) **Aug 2024 – Present**

Software Developer *New York, US*

- Architected an **NSF-funded ($100K)** AI-powered health app for **iOS** and **Android**, processing real-time wearable data; deployed cloud-hosted AI models to analyze health metrics and process food images.
- Engineered ETL pipelines transforming **NoSQL** into **PostgreSQL** using **concurrency control and locking mechanisms**; processed **250K+** records and powered **AWS QuickSight** dashboards for real-time health analytics.
- Developed an insurance analytics platform **(React, Node.js, REST APIs)**; set-up **(Docker** containers on **EC2**, implemented Stripe billing, RBAC, and row-level security, generating **$50K+** revenue.
- Automated underwriting via broker portal integrated with insurer platform; handling **document and media processing** and **WebSocket-based real-time communication**; which secured **6 insurance clients**.

PricewaterhouseCoopers (PwC) **Aug 2021 – Aug 2024**

Senior Software Engineer

- Orchestrated a distributed backend on **Microsoft Azure**, leveraging **Azure Blob Storage** and event-driven architecture to process large datasets, reducing data processing time by **75%**.
- Deployed an **NLP-based OCR neural model** on Azure to automate structured data extraction, saving the engineering team **15 hours of manual work per week**.
- Led code reviews and mentored engineers on **software design principles**, improving code quality and reducing bugs by **25%**.
- Optimized data ingestion pipelines using **parallel processing** for **100K-row Excel datasets**, reducing execution time from **5 hours to 1 hour** and enabling daily refreshed analytics.
- Customized **Power BI dashboards** processing large datasets from Polyglot databases, improving reporting efficiency by **40%**.

Projects

Hospital Management System (Full-Stack Application) | ***React, Node.js, Express, MongoDB, SageMaker, Hugging Face***

- Built a **full-stack healthcare platform** using **React, Node.js, Express, and MongoDB** to manage appointments, patient records, lab reports, and medical history with **role-based access control**.
- Implemented a **graph-based doctor recommendation system** using **Dijkstra's algorithm** to match patients with the nearest available doctor, improving booking efficiency by **45%**.
- Fine-tuned a **Hugging Face transformer-based LLM** achieving **88% accuracy** for health condition prediction and deployed it on **AWS SageMaker** for real-time diagnostic support in the doctor dashboard.

Skills

Programming Languages: Python, Java, C++, TypeScript, JavaScript (ES6), SQL
Web Technologies: REST API, GraphQL API, Microservices, React, Node.js, Express.js, Sequelize
Database: RDBMS (MySQL, PostgreSQL), NoSQL (Dynamo DB, FireBase, MongoDB), Dataverse
Cloud: AWS (AWS CDK, EC2, S3, API Gateway, Lambda, Sakemaker), Azure (OpenAI services, Function Apps, Storage blobs)
Tools: Git, Visual Studio, Cursor, Jupyter, Postman, Docker, Power Automate, Power BI, QuickSight
Methodology: Agile, Waterfall, Scrum
AI/ML: PyTorch, Hugging Face, LLM Fine-tuning, RAG (LangChain, FAISS), NLP, SVM, Random Forest, MLflow, PySpark

Certifications

AWS Cloud Practitioner
Microsoft Solution Architect Expert
Power BI Data Analyst Associate
Microsoft Developer Associate
Stanford Machine Learning (Coursera)
Azure Data Fundamentals